CS 285 Homework 5: Offline RL — Report

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All experiments use a single seed (`seed=0`) and the default hyperparameters shipped in the starter code (`hidden_size=256`, `num_layers=4`, `lr=3e-4`, `discount=0.99`, `target_update_rate=0.005`, `batch_size=256`; `flow_steps=10` for FQL; `expectile=0.9` for IQL). Each run is trained for 10^6 gradient steps with evaluations every 10^5 steps over 25 rollouts. All numbers below are the success rates produced by the starter-code evaluation harness.

1 SAC+BC (Section 2)

The implementation lives in `src/agents/sacbc_agent.py`.

Best-performing agents. Hyperparameter α was swept over $\{30, 100, 300, 1000\}$ on `cube-single` and over $\{1, 3, 10, 30\}$ on `antsoccer-arena`. The best run for each task is summarized in Table 1 and plotted in Figure 1.

Task	Best α	Final success @ 1M	Best success within 1M	Pass criterion
`cube-single`	100	**100%**	100%	> 75% final ✓
`antsoccer-arena`	10	12%	**44%**	> 5% ✓

Table 1: SAC+BC best agents.

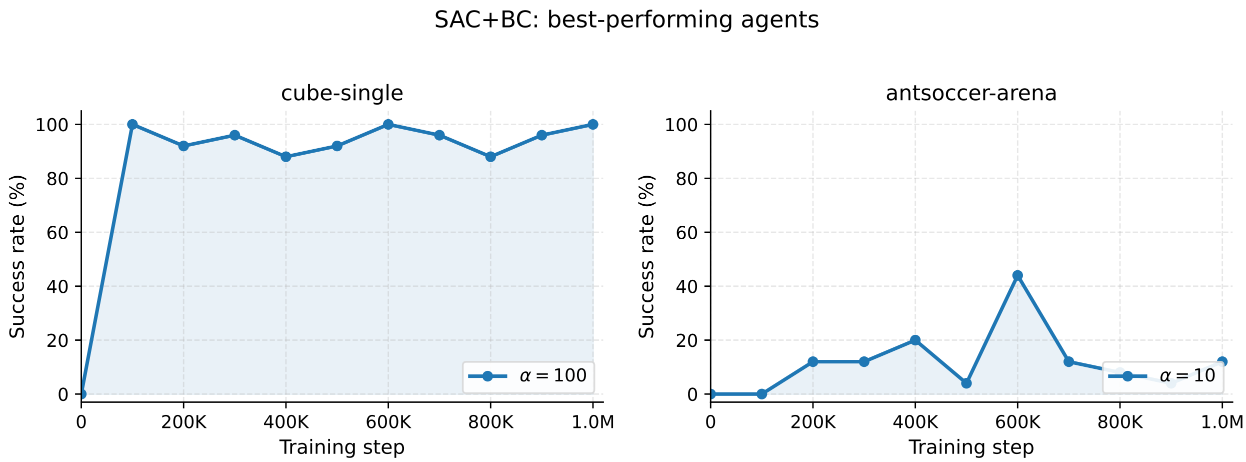

Figure 1: SAC+BC training curves of the best agents on `cube-single` ($\alpha = 100$) and `antsoccer-arena` ($\alpha = 10$).

Effect of α on `cube-single`. Figure 2 shows the four α values. Training is highly sensitive to α: the mid-range ($\alpha = 100, 300$) converges fast and stays near 100%, the small $\alpha = 30$ (BC term too weak) is unstable around 90%, and the large $\alpha = 1000$ (Q maximization swamped by BC) is slower to climb and noisier.

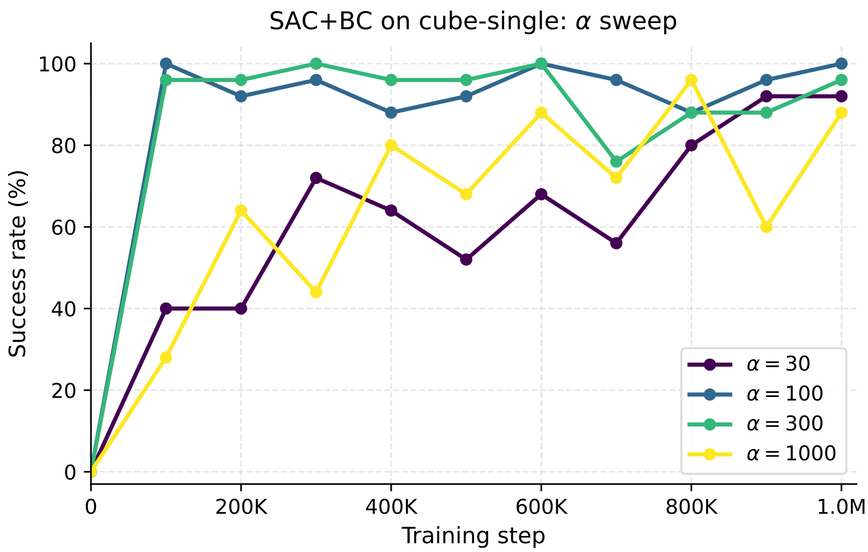

Figure 2: SAC+BC on `cube-single` with four BC coefficients.

2 IQL (Section 3)

The implementation lives in `src/agents/iql_agent.py`; the expectile parameter is fixed at $\tau = 0.9$ throughout.

Best-performing agents. Inverse-temperature α was swept over $\{1, 3, 10\}$ on both tasks. The best results are listed in Table 2 and plotted in Figure 3.

Task	Best α	Final success @ 1M	Best success within 1M	Pass criterion
`cube-single`	1	**88%**	100%	$> 60\%$ final ✓
`antsoccer-arena`	10	8%	**36%**	$> 5\%$ ✓

Table 2: IQL best agents (with $\tau = 0.9$).

Effect of α on `cube-single`. See Figure 4. All three α values quickly reach high success rate; $\alpha = 1$ converges to the highest plateau, while $\alpha = 3, 10$ converge to slightly lower but comparable plateaus.

Discussion: SAC+BC vs. IQL. (1) Best performance. SAC+BC is slightly stronger than IQL at the optimum on both tasks: on `cube-single` it reaches 100% final success versus IQL's 88% final (100% peak); on `antsoccer-arena` the within-1M peak is 44% versus IQL's 36%. **(2) Sensitivity to α.** IQL is markedly less sensitive: all three IQL $\alpha \in \{1, 3, 10\}$ land within ~20 percentage points of one another on `cube-single` and all yield non-trivial success on `antsoccer-arena`,

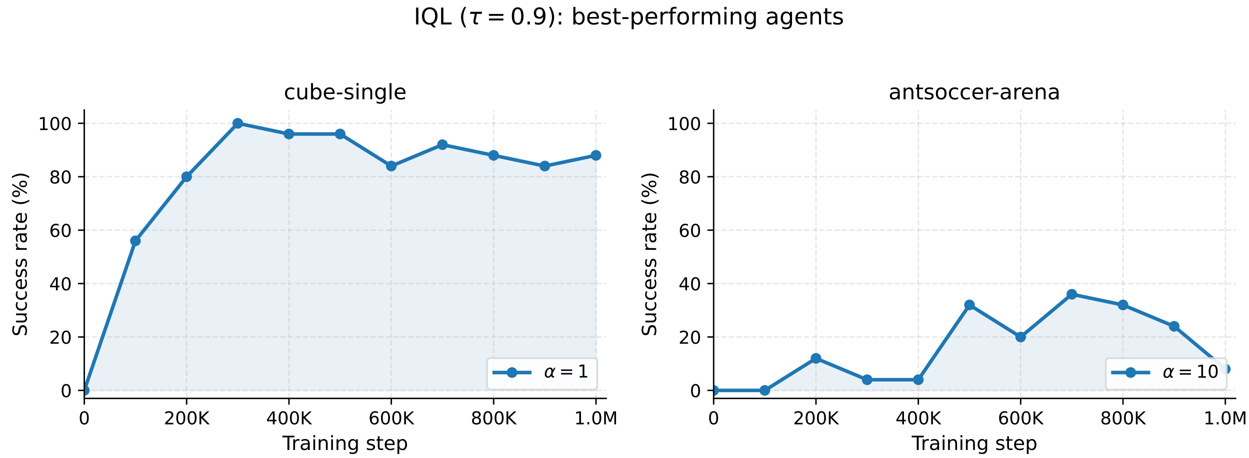

Figure 3: IQL training curves of the best agents on `cube-single` ($\alpha = 1$) and `antsoccer-arena` ($\alpha = 10$).

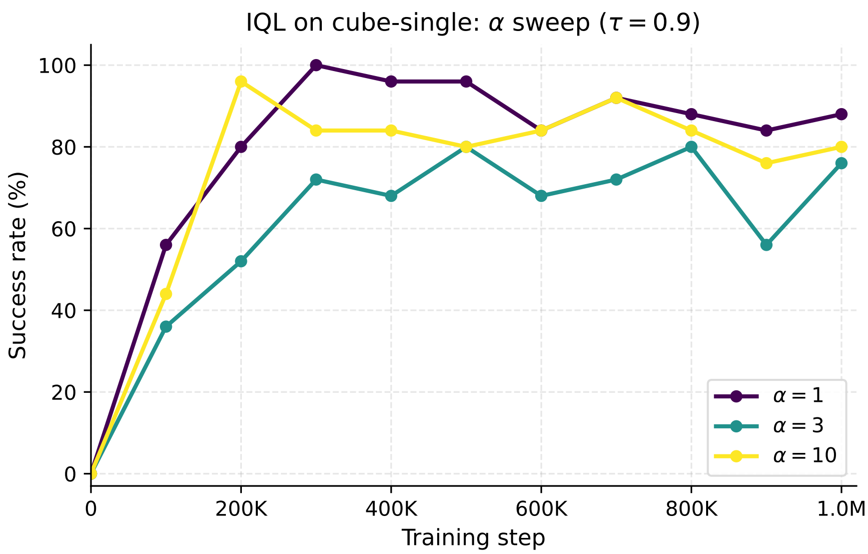

Figure 4: IQL on `cube-single` with three inverse temperatures.

whereas SAC+BC collapses to near-0% on `antsoccer-arena` when α moves to 30 and shows much wider spread on `cube-single`. This matches the textbook expectation that advantage-weighted regression makes IQL easier to tune.

3 FQL (Section 4)

The implementation lives in `src/agents/fql_agent.py`.

Best-performing agents. Distillation coefficient α was swept over $\{30, 100, 300, 1000\}$ on `cube-single` and over $\{1, 3, 10, 30\}$ on `antsoccer-arena`. The best run for each task is given in Table 3 and plotted in Figure 5.

Task	Best α	Final success @ 1M	Best success within 1M	Pass criterion
`cube-single`	300	**100%**	100%	$>$ 80% final ✓
`antsoccer-arena`	10	0%	**52%**	$>$ 30% best within 1M ✓

Table 3: FQL best agents.

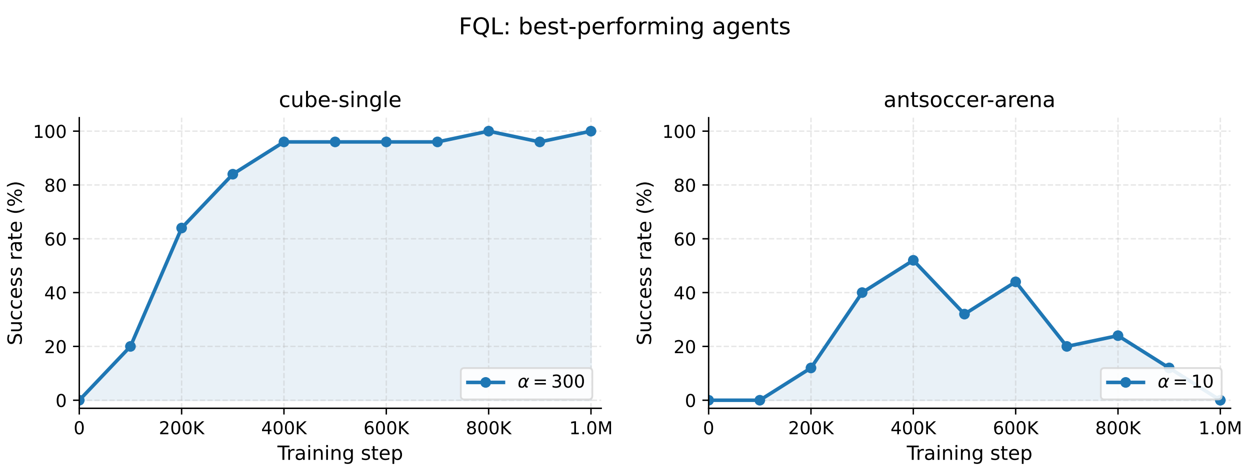

Figure 5: FQL training curves of the best agents on `cube-single` (α = 300) and `antsoccer-arena` ($\alpha = 10$).